Cluster 4: Fieldwork

1		A group of 4 students is interested to investigate if urban neighbourhoods in Singapore meet the needs of the elderly. They read an article about an upcoming project, the Toa Payoh Integrated Development, and want to investigate if it would supplement the Toa Payoh neighbourhood in meeting the needs of the elderly when it is completed. The students came up with the following hypothesis: "With the Toa Payoh Integrated Development, Toa Payoh will meet the needs of the elderly in the neighbourhood." The students went to Toa Payoh Hub to collect the data for the investigation every Monday from 8pm to 9pm for 4 weeks in August. To gather data from the elderly, the group administered a survey questionnaire in English. However, the group faced difficulties collecting the data as the elderly were cautious about them being scammers and some could not respond in English. The group ended up with a total of 15 responses. Resource 1 shows the map of Toa Payoh. Resource 2 shows the article on the upcoming Toa Payoh Integrated Development. Resource 3 shows the survey questionnaire that the students have crafted for the investigation. Resource 4 shows selected responses by the elderly to Question 10 of the survey.	
	(a)	Explain why the students' hypothesis is suitable for investigation.	[4]
		Award 1 mark for each explanation of the suitability of the students' hypothesis. Award 1 additional mark for the further elaboration of each point. Possible answers include: • [Clearly defined] There is a clearly defined population and variable with the group of people and what is to be studied about them. The population identified as the elderly living in Toa Payoh and the variable being the needs of elderly. • [Suitable scale] The hypothesis is of a suitable spatial scale of a neighbourhood. The hypothesis only investigates one neighbourhood in Singapore. • [Researchable and measurable] This hypothesis is capable of research with having 4 weeks to collect the data at Toa Payoh. ○ The students stationed themselves at Toa Payoh Hub which would have a regular flow of people allowing them to be able to collect data through their survey questionnaire.	

		• [Researchable and measurable] There are 4 students in this group, which allows them to split into two pairs to collect more responses at the same time. ○ The survey would help the students to understand the needs of elderly, hence it is capable of research. **AO1**	
	(b)	Using Resource 2, explain how Toa Payoh Integrated Development can help to meet the needs of the elderly.	[5]
		Award 1 mark for each explanation how the needs of the elderly can be met. Award 1 additional mark for the further elaboration of each point. Possible points include: • The improved sports facilities in the area can meet the elderly's psychological needs as it will be built for the "community to come together and bond through sporting activities" • Aimed to "help people achieve their health and wellness goals", the elderly can use these facilities to maintain or improve their health and fitness levels. ○ With the elderly being mobile and independent, they are able to meet their social and economic needs without having to rely on others. • The integrated development can help to meet the social needs of the elderly with the upgraded polyclinic placed in an accessible location to help them to seek "preventive health and early intervention" services. • The polyclinic will provide "senior services such as frailty screening and health promotion interventions to encourage healthy and active ageing". This helps to promote better health and allow the elderly to be active in the neighbourhood. • Toa Payoh Town Park can meet the psychological needs of the elderly with them being in nature more which can help to contribute "positively to mental resilience and improves well-being". **AO2**	
	(c)	Explain the ethical considerations the group should bear in mind when conducting surveys.	[4]
		Award 1 mark for each explanation point of ethical considerations the group should take. Award 1 additional mark for further elaboration of each point.	

		Possible answers include: • Ensure confidentiality of data collected, to inform the elderly the data collected will only be used for this research to ensure data privacy. • Inform respondents that their anonymity will be maintained and they will not be able to be identified. • Need to seek permission, or at least, notify gatekeepers, for conduct of research (e.g. Toa Payoh Hub Management) • Respondents must be given the autonomy to decide that they might not want to continue with the interviews at any point in time should they feel uncomfortable • Need for vetting and thinking through of questions, as well as pilot test, to ensure questions are not sensitive and would not unintentionally cause discomfort to the elderly. • Minimising impact of research on the communities: Reducing the amount of noise generated in an area, and not causing any damage nor inconveniences to the lives of those in the local community **AO1**	
	(d)	With reference to Resource 3, explain the strengths of the survey questionnaire that the students have crafted in providing data for this study.	[6]
		Award 1 mark for each explanation point of the survey questions' usefulness. Award 1 additional mark for further elaboration of each point. Possible answers include: • The survey questions are useful in collecting information about places the elderly like to visit and activities they enjoy in their free time. ○ This helps the students to infer if the hypothesis is true through comparing places the elderly frequent and their activities to the proposed facilities. • Students collecting data on age and ethnicity may aid in their investigation if they observe patterns among the different age groups and ethnicities. • The open-ended questions help to supplement and get more information from the close-ended questions. ○ This allows for deeper understanding of the needs of elderly in Toa Payoh • There are also questions that are relevant to the hypothesis regarding the needs of the elderly being met, this can be seen in Questions 7-9. ○ The data collected from this survey can show how the neighbourhood meets the elderly's needs.	

		AO2	
	(e)	Develop a plan on **<u>one</u>** other primary data collection method that can help to supplement the students' investigation.	[7]
		Award 1 mark for each step described to collect primary data. Award 1 additional mark for further elaboration of each point. Award a maximum of 2 marks for description of method, risks, ethics, limitations. Possible answers include: • [Method] Land use mapping, interview, focus group discussions • Students need to describe how the method can be conducted and how it helps to supplement the survey questionnaire that the students have • In their plan, students should also describe and explain the risks and ethical considerations as well as acknowledging the limitations of the method given <u>Example</u> Description of method • Students can use land use mapping to collect data on facilities and services provided for the elderly in the area. • Students can narrow down the area for the land use map by shortlisting a 200m radius around Toa Payoh Hub. o This can first be done with the reference of online maps, to help them to narrow down an area that can help to represent he neighbourhood. • To fill in the map, the students can walk along the stretch and record the different land uses of the area. Different land uses can be recorded with different initials or colours. o With this data, students can analyse the proportion of the area that meets the needs of the elderly. • Photographs of the area can be taken to help students to add in details about certain places in the map. Risks & Mitigations • While collecting data along the streets, students should avoid standing or walking on the roads to avoid accidents. • Students should be aware of surroundings when collecting data, to avoid injuries due to uneven pavements. Ethics • Students should avoid using the entire pavement, causing inconvenience to shoppers and commuters.	

		• Students should avoid leaving litter in the areas they conduct their landuse survey in. Limitations • Landuse mapping might require more time and visiting at night might not be a good representative of the elderly visiting the different shops and facilities. • The area shortlisted might not be an accurate representation of the neighbourhood as the other parts of the neighbourhood are largely residential. **AO1**	
	(f)	Using Resource 4, describe how the data collected from Question 7 of the survey can be analysed.	[4]
		Award 1 mark for each valid description. Award 1 additional mark for further elaboration of each point. Possible answers include: • Coding can be done to analyse data as shown in Resource 4 • Students can start by segmenting the answers into categories such as the different needs of the elderly. o The needs observed are mainly psychological and social needs. • The different needs of the elderly can be used as labels to identify patterns, meanings and themes. o Themes can include accessibility, activities to socialise and facilities to socialising • Students can do this manually or use a software to analyse the data. **AO2**	
	(g)	Based on the students' investigation, the students concluded that their hypothesis can be accepted. Evaluate the validity of the fieldwork.	[10]
		Possible approaches: Candidates could consider the validity of their hypothesis based on the data collected. They could refer to the Resources to see if the improvements that the elderly hope to have in Toa Payoh will be fulfilled with the new integrated development by referring to the information provided in Resource 2. Students could consider the way the investigation was conducted, including the data collection methods. They could analyse the amount of time spent at the investigation site along with the time of the day. They could also consider the	

		language used for the questionnaire and the target audience being senior citizens, leading to them facing issues in their data collection. For the data collected in Resource 4, candidates could consider the clarity of the terms used in the questionnaire survey and how these could give rise to differing interpretations by the elderly. For instance, to some elderly, satisfaction (e.g. Question 5 in Resource 3) can have different meanings to different individuals. Candidates could consider how such differing interpretations could affect the conclusions that could be drawn from the analysis of these data. Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions **AO3**	

Cluster 3: Sustainable Future and Climate Change

2	Resource 5 shows the percentage of urban slum population in developing regions in the world from 1990 to 2014. Resource 6 shows an excerpt on women's lives in slums. Resource 7 shows oxygen isotope data in ocean cores for the last 500,000 years. Resource 8 shows a cartoon on responses to climate change.		
	(a)	With reference to Resource 5, compare the percentage of urban slum population across developing regions in the world from 1990 to 2014. [5]	
		Award 1 mark for each valid comparison based on the trends observed in the resource. Award a maximum of 1 additional mark for further development of each comparison, where applicable. Possible answers include: • In general, all regions except West Asia and Oceania saw a decrease in percentage of urban slum population from 1990 to 2014 ○ For example, sub-saharan Africa saw a decrease from 70% to about 55%, while East Asia decreased from 45% to 27% • Only Oceania remained relatively the same at about 25%, while West Asia saw an overall increase of about 3% over the years • Sub-saharan Africa continues to have the highest urban slum population from 1990 to 2014 • West Asia had the least urban slum population in 1990 (23%) but North Africa had the least in 2014 (13%) • South Asia saw the largest decline in urban slum population of about 26% while South America and the Caribbean saw the smallest decline of about 12.6% • Most regions saw a gradual decline in urban slum population but 3 regions saw a more drastic decrease in urban slum population	

		○ South Asia, Southeast Asia and North Africa all saw a decline of at least 20% • West Asia's urban slum population pattern differed from the rest where it is the only region which experienced an increased in urban slum population during this time period. It declined from 1990 to 2000 but increased in 2005, before decreasing and increasing slightly again **AO2**
	(b)	Using Resource 6, explain the issues that are faced by women living in urban slums. [6]
		Award 1 mark for each explanation on issues that are more prevalently faced by women living in urban slums. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Psychological] Women living in urban slums may have to travel long distances to commute to and from work and when these public spaces are not designed with the needs of women in mind, they are exposed to greater danger and are exposed to higher risk of danger. Women 'suffer from security threats when they travel to and from their work on public transport', 'targeted for various kinds of assaults as they somehow come across as vulnerable and unprotected by the formal system of institutional support' ○ This instils fear in women, and the harassment they experience is also a threat to their psychological well-being • [Socio-economic] On top of having to work, women tend to spend disproportionate amount of time on unpaid work, such as 'attending to their care-giving roles at home'. The lack of access to basic services in urban slums increase the time they need to spend on such unpaid work ○ Women who spend a large proportion of their time completing unpaid care work, leaving them with little to no time left for paid employment, tend to have their economic well-being negatively affected. • [Economic] 'The neighbourhoods where these women live are often in slum areas, which are far from the places where they work every day'. This increases economic strain on women who may already have been receiving lower income due to the nature of job they are employed in ○ When urban poor women do find work, it is often precarious, and the wages are insufficient to escape poverty, affecting women living in urban slums socio-economically • [Social] Due to urban areas providing better access to employment opportunities, 'many young girls are pulled out of school and sent to work in the cities'. This result in the lack of access to formal education and increased gender inequality

		○ This has a knock-on effect where women tend to take up lower-skilled jobs that pay them less, which makes them less financially independent **AO2**
	(c)	Describe strategies implemented by cities to improve the lives of slum dwellers. [4]
		Award 1 mark for each explanation of a strategy to improve lives of slum dwellers. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Public Housing] Cities can relocate slum dwellers to new dwellings such as public housing that are built with permanent materials that can also withstand extremes in climate, as well as to provide slum dwellers with sufficient living space to reduce slum dwellers' risk of disease spreading and exposure to hazards ○ Often, such units come with basic infrastructure to adequately meet the essential needs of slum dwellers too • [Sites and Services Schemes] Sites and service schemes are land prepared with service infrastructure in place and either sold or leased, and new residents will build the house themselves or contract the work out. Key benefits of this scheme include security of tenure, adequate infrastructure, and freedom to build at one's own pace. ○ Sites and services schemes recognise the ability of people to house themselves with a little backing from the government agencies. As tenure is often more secure as it is given by the government, the beneficiaries would be more invested in improving their homes. It enables the low-income families to obtain decent housing and services, at levels that can be afforded by them • [Self-help housing] Self-help housing involves upgrading of existing squatter settlements via the provision of services and rationalisation of house and street layout; sometimes households are also provided resources to upgrade their own residence and serviced sites, sometimes with a partially completed unit, where the beneficiary is responsible for building or completing the unit. ○ States may provide legal title of land to households so that they will improve their houses without fear of eviction, as well as major infrastructure services and subsidised building materials for home improvements that improve their quality of living **AO1**
	(d)	Using Resource 7, explain how the study of ocean cores provides evidence of past climate variability. [4]

		Award 1 mark for each explanation on how the study of ice cores provides evidence of past climate variability. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Ocean cores] One proxy indicator is ocean cores that are drilled from the sea-floor to study the layers of shells that settle on the ocean bed over time. Seafloor sediments contain remains of organisms and when such organisms die, their shells slowly settle to the floor of the ocean. o Shells of tiny organisms, known as foraminifera, build calcareous shell around themselves and the chemical composition of the shells is affected by the chemistry of the ocean waters in which they live • [Oxygen isotopes] The ^{16}O isotope, which is slightly lighter, vaporises more readily whereas ^{18}O being heavier, condenses more easily. • [Resource] Resource 7 shows that as the percentage of ^{18}O in the ocean core increases, the earth enters glacial phase, whereas when the percentage of ^{18}O decreases as recorded in the ocean core, the earth enters interglacial phase o [Explanation] During glacial periods, ^{16}O molecules evaporates from the ocean and not returned via condensation (as they are 'locked' in snow), causing the remaining ocean water to become depleted in ^{16}O molecules (i.e. enriched with ^{18}O molecules) **AO2**
	(e)	Describe how the impacts of contemporary climate change vary. [6]
		Award 1 mark for each explanation on how impacts of contemporary climate change vary. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Nature of impact] Contemporary climate change can bring about both positive and negative impacts. o The increase in temperature can bring about increase in agricultural yields in temperate regions due to warmer weather during growing season, but droughts and heat-stress result in decline in agricultural yields in warmer regions of the world • [Uneven changes in temperature] Contemporary climate change results in variations in impacts experienced by places across the world due to uneven changes in temperature. Some places experience larger increase in temperature than others o Although contemporary climate change brings about an increase in global average temperature, the temperature change is larger over land

		than over the ocean due to differing specific heat capacity of land and ocean / higher latitudes are observed to experience larger increase in temperature • [Uneven changes in precipitation] Contemporary climate change results in variations in impacts experienced by places across the world due to uneven changes in precipitation across places – some locations experience increased precipitation and flooding, while others suffer from drought • [Risk to sea level rise] One of the impact of contemporary climate change is sea level rise and the risk to this impact vary based on exposure to the sea / altitude above sea level o Small island states are at high risk to sea level rise due to their countries being surrounded by the sea as compared to landlocked countries that face less of this impact • [Difference in vulnerability of different groups of people] Contemporary climate change poses disproportionate risks to humans due to differences in vulnerability and exposure to the impacts, resulting in the impacts of climate change to vary across social groups and communities o People who are older or younger are more vulnerable to climate-related health issues. People who have less economic resources have limited capacity to cope with the impacts of climate change, resulting in higher risk to the impacts. **AO1**
	(f)	With reference to Resource 8, explain the challenges in responses to climate change. [5]
		Award 1 mark for each explanation on the challenge in responding to climate change. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Role of firms / organization] As seen in Resource 8, the organization has decided to implement project where there is the overlap of 'easy to help' and 'most in need'. Organisations / firms have the financial resources to support climate change projects, but they may not necessarily focus on places most in need as they have to account for their investment, and may go with more visible projects that are easier to help and/or have tangible effects o As firms account to shareholders, when the interests of shareholders do not align with what the firms agreed upon in terms of climate action, firms may end up protecting the interests of their shareholders and their own firm over protecting the environment, and choose to invest in projects with more tangible impact • [Vulnerable communities] Countries suffering from the impacts of climate change are often countries at lower levels of development that require assistance to respond to climate change. Yet, Resource 8 shows that there is a lack of projects targeted at areas where they are most in need.

		○ Globally, there continues to be a lack of financing for CLLDs to implement projects to mitigate and adapt to climate change. Global climate fund set up at the international scale is still short of the target agreed • [Lack of stakeholder involvement] Successful planning and implementation of strategies need to be enhanced through complementary actions across levels, from individuals to governments. Resource 8 shows a lack of cooperation and collaboration across different stakeholders in addressing this transboundary issue ○ For example, Indigenous, local, and traditional knowledge systems and practices, including indigenous peoples' holistic view of community and environment, are a major resource for adapting to climate change, but these have not been used consistently in existing adaptation efforts. Integrating such forms of knowledge with existing practices would increase the effectiveness of adaptation. **AO2**

Cluster 3: Sustainable Future and Climate Change

3	'Urban reimaging is important in contributing to cities' progress towards sustainable urban development.' Evaluate this statement.	[20]
	Possible Approaches: Candidates could approach the question by making a judgement on whether urban reimaging strategies have been important in contributing to cities' progress towards sustainable urban development. Candidates could analyse the positive and negative impacts of the strategies in terms of the nature of impact (economic, social, environmental), spatial and temporal scale. Candidates could also bring into the discussion the complementarity and tension between liveability and sustainability, considering the importance of other aspects of liveability and sustainability that contribute to different dimensions of liveability. Candidates could also approach the question by making a judgement on whether urban reimaging strategies have been important in contributing to cities' progress towards sustainable urban development through a consideration of two or more case studies. Candidates could evaluate whether the strategies in each case study achieved the intended aims or targets. Candidates could also consider whether there are unintended negative impacts in achieving the aims or targets. Candidates could also analyse the challenges to success which are common across the case studies. *Levels marked using Generic Level Descriptors for 20m H2 essays* **AO3**	

4	'It is not possible to stop climate change.' Evaluate this statement.	[20]
	Possible Approaches: Candidates could approach the question by making a judgement on whether climate change can be stopped with a consideration of natural and human factors that contribute to climate change. Candidates may consider how climate change cannot be stopped due to natural factors (including variations in solar output, the thermohaline circulation, changes in ice sheet, volcanic activity and Milankovitch's cycle) that have resulted in past climate variability even before human existed, as well as human factors (such as the contribution to greenhouse gas sources and the removal of carbon sinks) that have exacerbated the rate climate is changing in contemporary times. Candidates could consider strategies humans implement to mitigate climate change along with the associated challenges. Candidate could also approach the question by making a judgement on whether human activities exacerbate or mitigate climate change. Candidates may consider how human activities (such as deforestation and industrial activities) can contribute to recent global warming, and weigh this against the extent to which human activities (such as reforestation or artificial carbon sequestration) may control or slow down the rate of climate change. Candidates could analyse this in terms of spatial scale, temporal scale, or the (in)adequate responsibility and commitment of relevant actors. *Levels marked using Generic Level Descriptors for 20m H2 essays* **AO3**	